Table 4: Performance on the AliMeeting and AISHELL-4 test sets with different training data, maximum chunk duration, and the use of word-level timestamps.

ID	Method	Word-level Timestamps	Training Data	Max Duration(s)	AliMeeting		AISHELL-4	
					cpCER (%)	tcpCER (%)	cpCER (%)	tcpCER (%)
M1	DM-ASR (Gemma3 270m)	✗	CN 212h	15	31.07	31.80	29.18	29.92
M2	DM-ASR (Gemma3 270m)	✓	CN 212h	15	28.24	28.96	27.75	28.51
M3	DM-ASR (Gemma3 270m)	✓	CN 630h	15	26.44	27.11	23.55	25.88
M4	DM-ASR (Gemma3 270m)	✓	CN 630h	25	24.33	24.98	22.03	23.74
M5	DM-ASR (Qwen3 0.6B)	✓	CN 630h	25	23.46	24.09	21.58	23.04
M6	DM-ASR (Qwen3 0.6B)	✓	CN 1300h	25	21.60	22.23	19.69	20.40

parameter-efficient adaptation, we fine-tune the model with LoRA while keeping the pretrained backbone frozen, and optimize the projector jointly. LoRA adapters are applied to both the speech encoder and the LLM, with scaling factor $\alpha = 32$ and rank $r = 16$. To obtain word-level timestamps, we use the MFA toolkit [33] to align the transcription with timestamps. During training, long-form recordings are segmented into 15-25 second clips. The model is trained on 8 NVIDIA A6000 48GB GPUs with a batch size of 2 per GPU, using the AdamW optimizer with a peak learning rate of 1×10^{-4} and a linear warmup-decay schedule.

4.3 Evaluation Metrics

We adopt the MeetEval3 evaluation protocol[6] and report three metrics to assess different aspects of multi-speaker transcription performance.

Diarization Error Rate (DER) measures the quality of speaker attribution by accounting for speaker confusion, missed speech, and false alarms. In our evaluation, DER is computed without a forgiveness collar, making it a strict measure of how accurately the system determines who speaks when.

Concatenated minimum-Permutation WER/CER (cpWER/cpCER) evaluates speaker-attributed transcription under permutation invariance. It concatenates all utterances assigned to the same speaker and computes the minimum error rate over all possible speaker permutations, which captures both lexical accuracy and speaker consistency.

Time-Constrained minimum-Permutation WER/CER (tcpWER/tcpCER) extends cpWER/cpCER by enforcing temporal constraints, such that words or characters are only matched within a predefined temporal collar. This makes the metric sensitive not only to speaker assignment and recognition accuracy, but also to timestamp quality, thereby jointly evaluating who, what, and when.

All metrics are reported in percentage (%), where lower values indicate better performance. For comparison, we include three types of baselines: Cascade Baselines; End-to-End Baselines, including large multimodal foundation models (Qwen2.5-Omni-7B [52], Gemini-2.5-Pro[7] , and Gemini-3-Pro[8]) and recent speech LLMs (SpeakerLM [54], JEDIS-LLM [45], Tagspeech [21], MOSS Transcribe Diarize [55], and VibeVoice-ASR [42]); Semi-end-to-end MS-ASR(Diarization-aware) baseline [31].

5 Experiment Results

5.1 Main Results

5.1.1 Results on Mandarin Multi-speaker ASR. Table 2 compares our model with strong cascaded baselines, recent end-to-end systems, and a semi-end-to-end baseline on the AliMeeting and AISHELL-4 test sets. Here, **DM-ASR (DiariZen)** uses diarization labels produced by DiariZen as front-end cues, while **DM-ASR (S2SND)** uses labels generated by an S2SND-style front-end whose original Conformer encoder is replaced with w2v-BERT 2.0 [10, 28]. Neither variant relies on oracle VAD information.

Overall, DM-ASR achieves clearly stronger structured multi-speaker transcription performance than the compared baselines. Relative to cascaded systems, it consistently yields much lower cpCER and tcpCER, showing the advantage of integrating diarization cues directly into generation rather than treating diarization and ASR as loosely connected stages. Compared with large multimodal models such as Gemini-2.5-Pro, Gemini-3.0-Pro, and Qwen2.5-Omni-7B, our method also achieves better DER, cpCER, and tcpCER in nearly all settings, despite using a much smaller model. This suggests that, for structured multi-speaker ASR, explicitly modeling speaker and temporal information is more effective than relying only on the implicit capabilities of general-purpose multimodal models. DM-ASR further outperforms recent speech LLM and semi-end-to-end baselines. In particular, compared with SpeakerLM trained on a similar data scale, DM-ASR remains clearly better, and even compared with SpeakerLM under its best reported setting, our method is still competitive(AISHELL-4 test sets). This suggests that the gains come not only from more data, but also from the proposed diarization-aware design itself.

Within our method, several clear trends can also be observed. Increasing the training data generally leads to better results, with the most consistent gains on cpCER and tcpCER, suggesting that more data helps improve speaker consistency and temporal modeling. Under comparable training conditions, the larger model generally

[6] https://github.com/fgnt/meeteval
[7] https://ai.google.dev/gemini-api/docs/models?hl=zh-cn#gemini-2.5-pro
[8] https://ai.google.dev/gemini-api/docs/models?hl=zh-cn#gemini-3-pro

outperforms the smaller one, indicating that stronger model capacity can exploit diarization-aware cues more effectively. Replacing DiariZen with the stronger S2SND front-end further improves DER and usually also improves cpCER and tcpCER, confirming the importance of front-end diarization quality. Adding bilingual Chinese-English training data consistently improves AISHELL-4, but does not consistently benefit AliMeeting and can even degrade performance there in some settings, suggesting that multilingual training mainly enhances cross-condition generalization rather than yielding uniform gains across all test sets. Overall, data scale, model capacity, and multilingual training are complementary, with the first two providing more stable improvements and the last being more dataset-dependent.

5.1.2 Results on English Multi-speaker ASR. Table 3 reports the results on the English test sets, we can observe that DM-ASR achieves strong and consistent performance across all English benchmarks, demonstrating that the proposed diarization-aware multi-turn framework generalizes well to English multi-speaker ASR. And although the DiariZen front-end was not trained on near-field data and therefore performs less favorably on DER for AMI-IHM, DM-ASR based on DiariZen-predicted labels still delivers strong multi-speaker ASR results. This indicates that our framework can effectively exploit imperfect diarization cues rather than being dominated by front-end errors.

Moreover, replacing DiariZen with the improved S2SND-based front-end leads to further gains while maintaining strong performance on AMI-IHM. This shows that our method is stable across different front-end diarization systems: stronger diarization cues bring further improvements, but the proposed framework itself remains effective even when the front-end is less matched to the target condition.

Finally, compared with JEDIS-LLM, DM-ASR achieves better AMI-IHM performance and comparable Fisher performance with a much smaller model and substantially less training data. This comparison further demonstrates the effectiveness of the proposed diarization-aware design, showing that strong English multi-speaker ASR performance can be achieved without relying on large-scale model or data scaling. Among all our variants, **DM-ASR (S2SND) (CN+EN 2900h) with a 1.7B LLM** achieves the best overall performance.

5.2 Ablation Study

Table 4 reports the ablation results on AliMeeting and AISHELL-4 under different settings of word-level timestamp supervision, training data scale, maximum chunk duration, and model size.

Comparing M1 and M2 shows that word-level timestamp supervision consistently improves performance on both datasets, indicating that fine-grained temporal grounding benefits both structured prediction and transcription quality. This trend is further supported by M2–M3 and M5–M6, where increasing the training data scale yields clear gains. A similar effect is observed from M3 to M4: extending the maximum chunk duration from 15s to 25s also improves performance, likely because longer chunks provide richer conversational context. On top of that, M4 and M5 show that scaling the model from 270M to 0.6B brings additional gains under the same data and chunk settings. Overall, word-level timestamp supervision, more training data, longer chunk duration, and larger model size all contribute positively to performance.

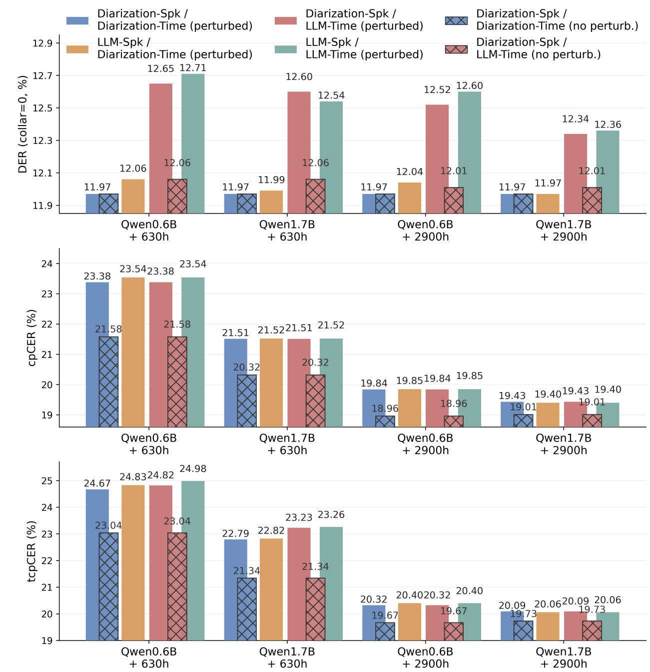

Figure 3: Performance comparison across different evaluation setups on AISHELL-4, measured by DER, cpCER, and tcpCER, under perturbed(spk/time,p=0.1) and non-perturbed training conditions.

5.3 Different Evaluation Setups

Figure 3 compares different evaluation setups on AISHELL-4 in terms of DER, cpCER, and tcpCER under both perturbed and non-perturbed training conditions. Several clear trends can be observed. First, using diarization-provided speaker and time information consistently yields the best performance, especially on DER, showing that reliable external diarization cues remain highly beneficial for speaker attribution. Second, the degradation caused by replacing diarization cues with LLM predictions is relatively limited, which suggests that the model can exploit contextual information to partially correct imperfect speaker and temporal information rather than simply depending on the prompt. Third, increasing the training data scale leads to substantial and consistent improvements across DER, cpCER, and tcpCER, in many cases exceeding the gains obtained by scaling model size alone. Fourth, the gap between different evaluation setups gradually narrows as both model size and training data scale increase. Finally, compared with the Qwen0.6B+630h setting, larger and better-trained models show more stable performance across all setups, particularly when both speaker labels and timestamps are predicted by the LLM, indicating stronger robustness and better cue refinement ability. This trend is also reflected in higher-resource settings (1.7B+2900h), where the benefit of perturbed training lies mainly in improving robustness under imperfect diarization cues, rather than further improving the best-case setup.

6 Conclusion

In this paper, we proposed DM-ASR, a diarization-aware multi-speaker ASR framework that reformulates multi-speaker transcription as a multi-turn dialogue generation problem. By explicitly introducing diarization cues into structured generation, DM-ASR enables a compact model to perform speaker-aware and temporally grounded transcription more effectively. Experiments on both Mandarin and English benchmarks show that DM-ASR achieves strong multi-speaker ASR performance with a relatively small model and limited training data. The results also demonstrate that word-level timestamp prediction not only provides richer structured outputs, but also improves transcription quality. In addition, analyses under different evaluation setups show that the model can effectively exploit reliable diarization cues and, with increased model and data scale, further correct imperfect speaker and timestamp conditions.

Overall, our findings suggest that explicit diarization information remains a valuable structural prior for small-size multi-speaker ASR models, even in the era of unified Speech-LLMs. At the same time, our results also reveal an important limitation: under relatively small model and data scales, fully LLM-predicted speaker labels and timestamps still do not consistently outperform strong diarization front-ends. We believe this gap can be further reduced, and potentially reversed, by scaling up both model capacity and training data, which will be an important direction for future work.

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